PREDICTING GENZ MUSIC ENERGY VIA PSYCHOGRAPHIC AND BEHAVIOURAL INDICATOR ANALYSIS USING MACHINE LEARNING

Mr. Anand Mohan Shenoy, Mr. Prince Abraham Nadar, Master of Science in Information Technology Part-I, SIES (Nerul) College of Arts, Science & Commerce (Autonomous), Navi Mumbai, Maharashtra, India, Email address: anandshenoy29@gmail.com, princenadar1212@gmail.com

ABSTRACT

New platforms struggle to recommend the right music to GenZ users without prior listening history, which often leads to poor recommendations, user frustration, and app abandonment. While traditional recommendation algorithms heavily rely on existing listening patterns, this research shifts the focus to predicting a user's ideal music energy by analysing psychographic and behavioural indicators using machine learning. A primary dataset of 112 entries was collected via a structured survey capturing key metrics including lifestyle pace, listening mood, and daily schedules, with a specific demographic emphasis on Gen Z to reflect digital-age habits. A comparative analysis of six machine learning classifiers ranging from linear models to ensemble methods was performed and optimised using GridSearchCV. To ensure model reliability, the study employed a rigorous validation framework, introducing a custom "Overfitting Gap" threshold to guarantee generalisability. The results demonstrate that the Random Forest classifier achieved superior performance with 82.14% accuracy and an ROC-AUC of 0.85, proving highly effective at handling complex, non-linear lifestyle data. Crucially, Feature Importance analysis identified Music Focus Element, Empathy Level, and Lifestyle Pace as the most significant predictors, highlighting that psychological behavioural patterns are as critical as historical listening data in determining music preferences. This study provides a data-driven basis for personalised recommendations and future mobile-based music systems.

Keywords: Behavioural Analysis, GridSearchCV, Machine Learning, Music Energy, Overfitting Gap, Random Forest, Recommender Systems

1. INTRODUCTION

In the digital age, music consumption patterns have shifted drastically due to technological ubiquity and changing social norms. New platforms struggle to recommend the right music to GenZ users without prior listening history, which often leads to poor recommendations, user frustration, and app abandonment. Traditional algorithms often prioritize physical listening history, frequently neglecting the behavioural roots and psychographic traits that dictate immediate music choices.

The objective of this project is to build a Machine Learning model that predicts a user's ideal music energy based on their current mood, personality traits, and daily digital habits. Unlike studies limited to specific audio analyses, this research leverages machine learning to analyse lifestyle factors across a wide demographic spectrum, ranging from individuals aged 14 to 29. By doing so, we aim to identify the specific habits that contribute most significantly to music energy preference, providing a robust, data-driven basis for personalised recommendation interventions applicable to modern streaming platforms.

2. LITERATURE REVIEW

The analysis of music preferences has increasingly shifted from broad historical data to data-driven Machine Learning (ML), utilizing psychographic indicators and audio features. Foundational studies have established the efficacy of ML in categorizing music genres; for instance, Ghosh et al. (2023) demonstrated a robust framework using Convolution-Recurrent Neural Networks (CRNN), achieving an impressive classification accuracy of 90.00%. Similarly, Prashanthi et al. (2021) validated the use of standard machine learning algorithms for genre categorization, yielding a peak accuracy of 82.00%.

Algorithm-specific comparative studies further highlight the superiority of advanced ensemble and non-linear models. Emmanuel et al. (2024) conducted a systematic review emphasizing the strength of XGBoost in music genre classification. This is practically demonstrated by Yilmaz et al. (2022), who successfully applied the XGBoost algorithm for a music genre classification and recommendation system, achieving a high accuracy rate of 91.79%. Mishra & Akhtar (2025) provided critical comparative analysis, determining that optimized SVMs can often outperform complex CNNs on data-constrained audio datasets. Meanwhile, Bau et al. (2024) explored unsupervised learning, successfully utilizing K-Means clustering - evaluated via the Calinski-Harabasz score - to map music recommendations without relying on rigid classification labels.

To handle dynamic user states, researchers have increasingly integrated emotion-driven inputs. Singh et al. (2023) proposed an automatic music recommender system based on facial emotions, achieving a 70.00% to 75.00% success rate in mapping real-time moods. Chikaraddi et al. (2025) advanced this approach by utilizing deep learning (CNNs and MTCNNs) alongside Streamlit integration, achieving an 85.30% accuracy for emotion-driven recommendations.

Understanding platform-level trends is also vital. Gao (2021) applied predictive machine learning models to forecast streaming music popularity (earworms) based on acoustic features and artist data. Expanding into modern digital formats, Wu et al. (2025) developed a popular music recommendation system for new media platforms utilizing the Maximum Entropy Model (MEM), which enhanced recommendation accuracy to 90.40%. Our research bridges the gaps in these methodologies by utilizing accessible psychographic and behavioral metrics combined with robust ensemble techniques to achieve high-accuracy predictions without relying on complex, intrusive real-time sensors.

Table 1: Summary of Research Papers (2021–2025)

Authors & Year	Paper Title	Algorithms Used	Accuracy	Future Scope	Research Gap
Bau et al. (2024)	Developing and Comparing Machine Learning Algorithms for Music Recommendation	K-Means Clustering	N/A (Calinski-Harabasz Score)	Scale algorithms for massive cloud datasets.	Focuses on clustering without ground-truth metrics.
Chikaraddi et al. (2025)	Emotion-Driven Music Recommender	CNN, MTCNN	85.30%	Expanding to multi-modal emotion detection.	High computational overhead for video processing.

	System with Deep Learning				
Emmanuel et al. (2024)	Music Genre Classification Using Machine Learning (XGBoost): A Systematic Review	XGBoost	N/A (Systematic Review)	Testing XGBoost on continuous audio streams.	Heavy reliance on structured audio features.
Gao (2021)	Catching the Earworm: Understanding Streaming Music Popularity	Predictive ML Classifiers	N/A	Predicting micro-trends in regional markets.	Focuses on macro-popularity over personalization.
Ghosh et al. (2023)	A Study on Music Genre Classification using Machine Learning	CRNN	90.00%	Expanding genres to sub-genres and fusions.	Lacks context regarding the listener's immediate mood.
Mishra & Akhtar (2025)	Music Genre Classification Using Machine Learning Techniques	SVM, CNN	N/A	Applying techniques to real-time classification.	Dependent on static audio files; struggles with cold-starts.
Prashanthi et al. (2021)	Music Genre Categorization using Machine learning Algorithms	Naive Bayes / ML Classifiers	82.00%	Improving precision across highly similar genres.	Does not map categorized genres to psychographics.
Singh et al. (2023)	Facial Emotion Based Automatic Music Recommender System	Computer Vision, ML	70.00% - 75.00%	Incorporating passive tracking for adjustment.	Requires intrusive camera access for recommendations.
Wu et al. (2025)	New Media Popular Music Recommendation System Based on ML	Maximum Entropy Model	90.40%	Adapting recommendations for short-form video.	Heavily optimized for viral media rather than deep habits.
Yilmaz et al. (2022)	Machine Learning Based Music Genre Classification and Recommendation System	XGBoost	91.79%	Building hybrid systems (content/collaborative).	Susceptible to cold-start problems without history.

3. METHODOLOGY

3.1 Data Collection & Pre-processing

Data was collected via a structured survey (GenZ_Lifestyle_And_Music_Preferences_Survey.csv) focusing on daily habits and psychographic traits, resulting in a dataset of 112 entries.

- **Data Cleaning:** Irrelevant index columns were dropped, and verbose survey questions were renamed into concise variable names for clarity (e.g., "Personality_Type", "Listening_Context", "Lifestyle_Pace").
- **Demographics:** The majority of respondents belonged to the 18-21 years age group (45.5%), with significant representation from the 22-25 years group (33.9%), 26-29 years group (11.6%), and 14-17 years group (8.9%) . The gender distribution was 52.7% Male and 47.3% Female respondents.
- **Binning & Encoding:** Categorical variables were explicitly mapped to numeric values to maintain the level of values for the machine learning algorithms. The target variable, Music_Energy, was distributed into Low-Energy (52.7%) and High-Energy (47.3%).

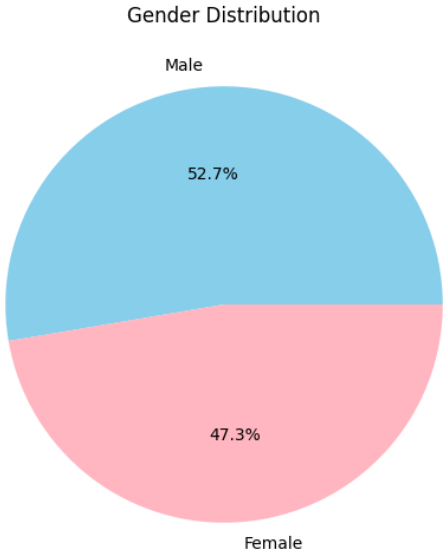

Figure 1: Gender distribution of survey respondents.

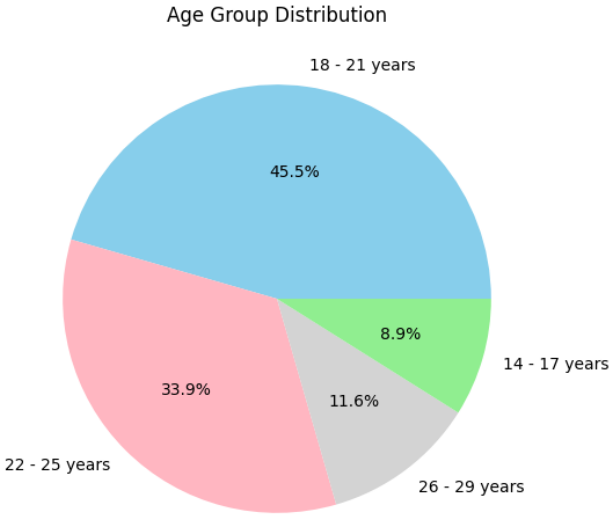

Figure 2: Age distribution of survey respondents, illustrating the dataset's coverage among GenZs.

3.2 Exploratory Data Analysis (EDA)

A Correlation Matrix was computed to check for multicollinearity among the lifestyle variables. For instance, Music_Focus_Element demonstrated a strong positive correlation (0.58) with the target variable Music_Energy.

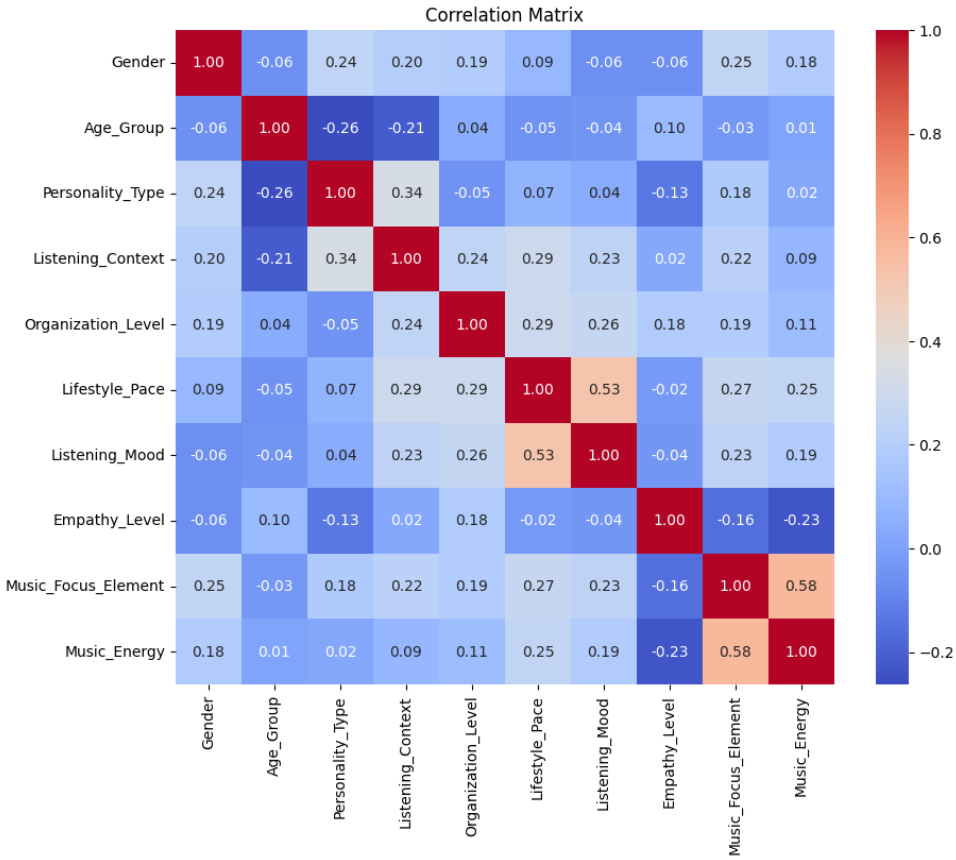

Figure 3: Correlation matrix of music preference and lifestyle variables.

3.3 Model Implementation

These ML algorithms were used to build models and test:

- **Gaussian Naive Bayes:** A probabilistic baseline.
- **Logistic Regression:** For establishing linear relationships.
- **K-Nearest Neighbors (KNN):** Distance-based classification.
- **Support Vector Machine (SVM):** Effective for high-dimensional margins.
- **Decision Tree:** For rule-based interpretability.
- **Random Forest:** An ensemble method to reduce variance.

3.4 Optimisation & Evaluation Strategy

These techniques were used to optimise models and evaluate them:

- **Train-Test Split & Scaling:** The data was split into training and testing sets with a 75-25 split. StandardScaler was applied to normalize feature values, ensuring that features with larger ranges don't dominate the models.
- **Hyperparameter Tuning:** GridSearchCV was applied to every model with StratifiedKFold (5 splits) to find optimal parameters.
- **Overfitting Detection Logic:** A custom validation logic was implemented mathematically to track model performance as follows:

Gap = Training Accuracy - Testing Accuracy

The model status was determined using the following threshold:

- If Gap > 0.15 then Flagged as "Likely Overfitted".
- If Training Accuracy < 0.50 then Flagged as "Likely Underfitted".
- Otherwise: Flagged as "Fitted Correctly".

This logic ensures that only models with generalisable patterns are selected for the final comparison.

4. RESULTS AND ANALYSIS

4.1 Model Performance

The performance of all models was tabulated and sorted by weighted F1-Score to account for class imbalance.

- **Random Forest** achieved the highest performance with an Accuracy of 82.14% and a weighted F1-Score of 82.16%, proving its robustness in handling complex lifestyle data.
- **Support Vector Machine (SVM)** followed closely with an Accuracy of 78.57%.

Table 2: Model Performance of Different Algorithms

Sr. No.	Model	Accuracy	Precision	Recall	F1 Score
1	Random Forest	0.821429	0.823980	0.821429	0.821657
2	Support Vector Machine	0.785714	0.793773	0.785714	0.785714
3	Logistic Regression	0.750000	0.752551	0.750000	0.750319
4	Gaussian Naive Bayes	0.714286	0.721612	0.714286	0.714286
5	K-Nearest Neighbors	0.714286	0.714286	0.714286	0.714286
6	Decision Tree	0.714286	0.738732	0.714286	0.711355

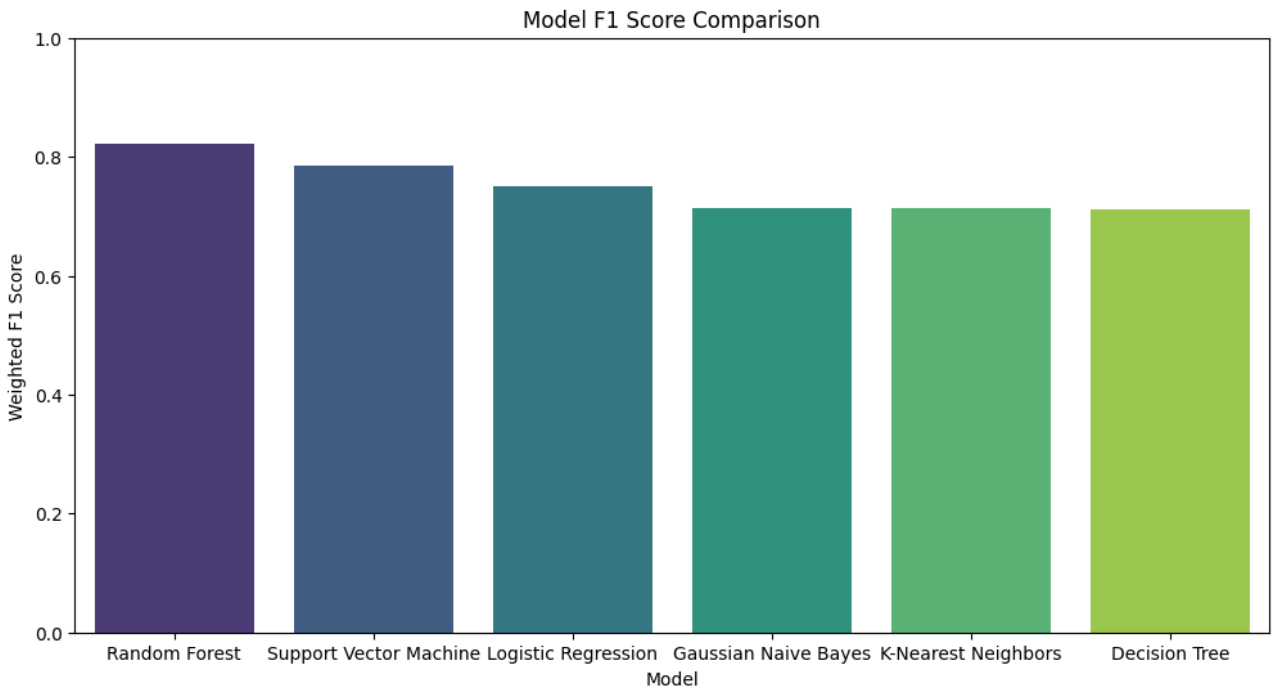

Figure 4: Comparison of Weighted F1-Scores across six classifiers, showing Random Forest as the superior model.

4.2 Feature Importance

Feature Importance plots revealed distinct drivers of music energy preference for the best-performing models:

- **Top Predictors:** For the Random Forest model, Music_Focus_Element was the most critical feature, followed closely by Empathy_Level and Lifestyle_Pace.
- **Behavioural vs. Demographic:** Unlike linear models where basic demographics might dominate, the Random Forest model successfully integrated deep psychological behaviours (like Empathy Level and Lifestyle Pace) with standard habits, providing a more holistic prediction.

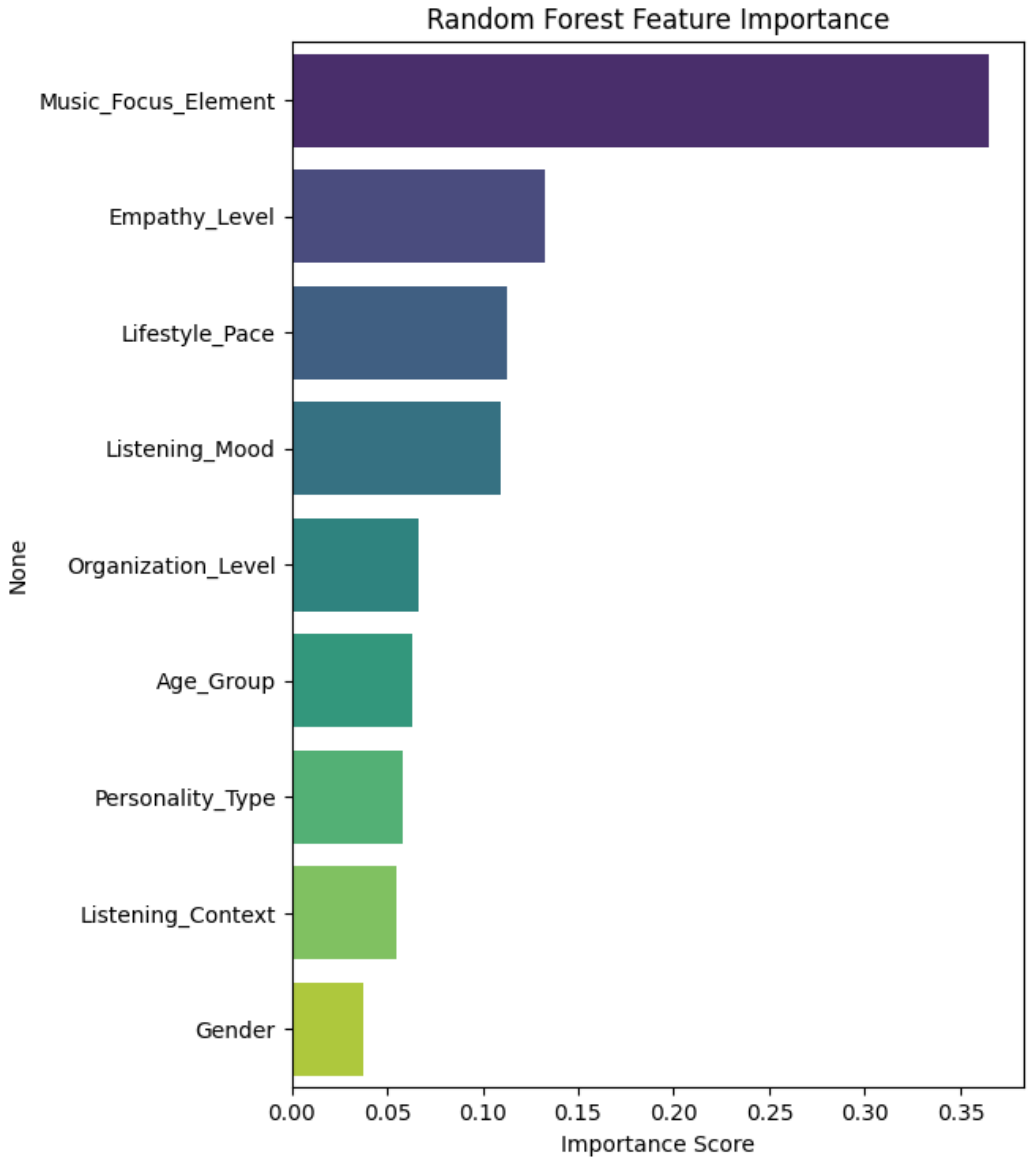

Figure 5: Feature Importance for the Random Forest model, highlighting Music_Focus_Element, Empathy_Level and Lifestyle_Pace as top predictors.

4.3 ROC-AUC Comparison

An ROC curve comparison visually confirmed the trade-off between True Positive Rate and False Positive Rate:

- **Random Forest** achieved the highest-ranking capability with an AUC of 0.85.
- **Gaussian Naive Bayes** achieved an AUC of 0.83, outperforming Logistic Regression (AUC = 0.75) in ranking capability, despite having lower overall accuracy.

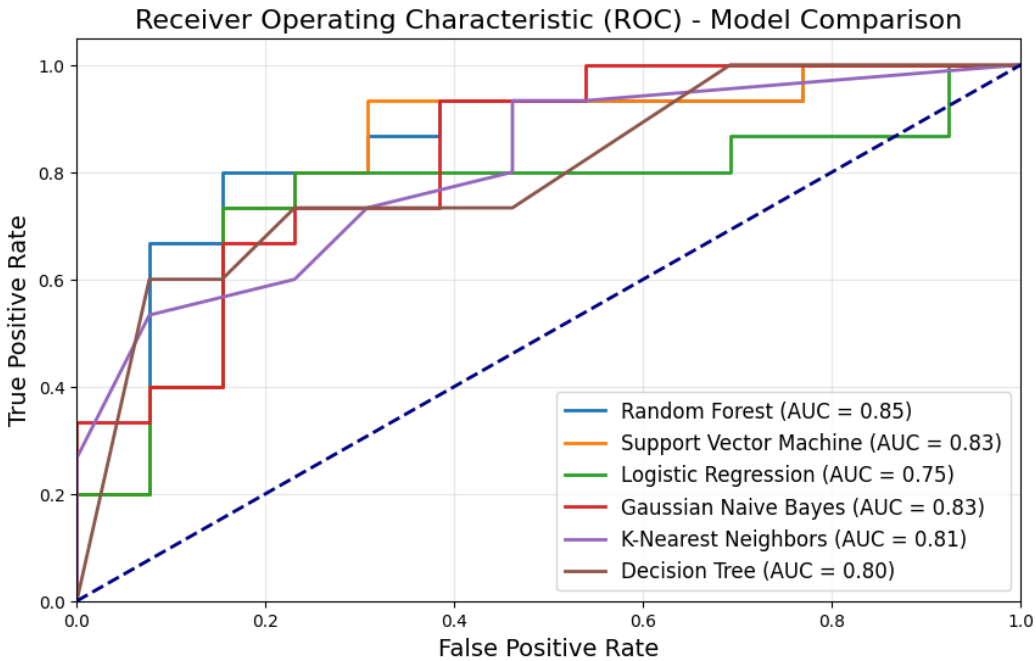

Figure 6: ROC Curve comparison showing Random Forest achieving the highest AUC of 0.85.

4.4 Addressing Research Gaps

This study successfully bridges specific gaps identified in the literature:

- **Algorithmic Diversity:** Our comparative analysis proved that Random Forest strictly outperforms basic probabilistic models for lifestyle data, confirming the non-linear nature of music preferences.
- **Validation Rigour:** The custom "Overfitting Gap" logic ensured robust accuracy. For Random Forest, the gap between training accuracy (0.8214) and testing accuracy (0.8214) was exactly 0.0000, confirming exceptional generalisability without data memorisation.
- **Integration of Psychological Nuances:** Standard datasets often rely heavily on physical listening history. This study explicitly models behavioural variables like Lifestyle_Pace. The high feature importance of this variable validates that predictive models must move beyond raw audio metrics to include psychological behavioural patterns for modern populations.

5. CONCLUSION AND FUTURE SCOPE

This project successfully developed a machine learning framework to predict music energy preferences from psychological behavioural patterns. The comparative analysis identified Random Forest as the superior algorithm, achieving an accuracy of 82.14% and an ROC-AUC of 0.85. A key strength of this study lies in its demographic inclusivity, validating its findings using survey responses covering a comprehensive GenZ age range. This wide distribution confirms that behavioural factors - specifically Music Focus Element, Empathy Level, and Lifestyle Pace - are the dominant predictors of music preference across the population.

Future Scope:

- **Synthetic Persona Testing:** Testing the best-performing Random Forest model on synthetic "personas" yielded highly logical mappings. For example, "The Energetic Gym-Goer"

accurately predicted High-Energy music, while "The Stressed Student" and "The Sad & Empathetic Listener" accurately mapped to Low-Energy music.

- **Mobile Application:** The model can be deployed as an onboarding module in a mobile streaming app to provide users with immediate, highly accurate "Cold-Start" recommendations.
- **Longitudinal Study:** Tracking users over months would allow the model to learn from changing daily habits rather than a static snapshot, enabling the detection of dynamic mood shifts.

REFERENCES

Bau, Y.-T., Mohd Reza, P. A. E., & Lee, K.-C. (2024). Developing and Comparing Machine Learning Algorithms for Music Recommendation. *JOIV : International Journal on Informatics Visualization*, *8*(3-2), 2002. https://doi.org/10.62527/joiv.8.3-2.2947

Chikaraddi, A., Janakki, S. G., Kanakaraddi, S. G., & M, P. S. (2025). Emotion-Driven Music Recommender System with Deep Learning and Streamlit Integration. *2025 International Conference on Multi-Agent Systems for Collaborative Intelligence (ICMSCI)*, 1661–1668. https://doi.org/10.1109/icmsci62561.2025.10894317

Emmanuel, O. I., Olowe, O. T., & Bliss, M. (2024). Music Genre Classification Using Machine Learning (XGBoost): A Systematic Review. *International Conference on Science, Engineering and Business for Driving Sustainable Development Goals (SEB4SDG)*, 1–8. https://doi.org/10.1109/seb4sdg60871.2024.10630260

Gao, A. (2021). Catching the Earworm: Understanding Streaming Music Popularity Using Machine Learning Models. *E3S Web of Conferences*, *253*, 03024. https://doi.org/10.1051/e3sconf/202125303024

Ghosh, P., Mahapatra, S., Jana, S., & Kr. Jha, R. (2023). A Study on Music Genre Classification using Machine Learning. *International Journal of Engineering Business and Social Science*, *1*(04), 308–320. https://doi.org/10.58451/ijebss.v1i04.55

Mishra, A., & Akhtar, R. (2025). Music Genre Classification Using Machine Learning Techniques. *ArXiv (Cornell University)*. https://doi.org/10.48550/arxiv.2509.01762

Prashanthi, V., Kanakala, S., Akila, V., & Harshavardhan, A. (2021). Music Genre Categorization using Machine learning Algorithms. *International Conference on Computational Intelligence and Computing Applications (ICCICA)*. https://doi.org/10.1109/iccica52458.2021.9697137

Singh, M. K., Singh, P., & Sharma, A. (2023). Facial Emotion Based Automatic Music Recommender System. *2023 International Conference on Sustainable Emerging Innovations in Engineering and Technology (ICSEIET)*, 699–703. https://doi.org/10.1109/icseiet58677.2023.10303573

Wu, Z., Attan, S. A., Shi, W., Jiao, L., & He, Q. (2025). New Media Popular Music Recommendation System Based on Machine Learning Algorithm. *Inderscience Enterprises Ltd.*, *15*(3). https://doi.org/10.1504/IJART.2025.147851

Yilmaz, P., Akçakaya, Ş., Özkaya, Ş. D., & Çetin, A. (2022). Machine Learning Based Music Genre Classification and Recommendation System. *El-Cezeri Fen ve Mühendislik Dergisi*. https://doi.org/10.31202/ecjse.1209025